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An Ensemble Machine Learning Framework for Accurate Prediction of Prostate Cancer from Clinical Data

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***Abstract*—Prostate cancer ranks among the most destructive and fatal illnesses for men globally, taking numerous lives annually. Despite considerable progress in medical technology and diagnostic techniques, reliably detecting the disease in its early stages continues to be a significant hurdle. PSA tests and biopsies represent traditional testing methods that often produce inaccurate or postponed results, hindering treatment and increasing mortality rates. This research introduces a successful machine learning framework to enhance the precision of prostate cancer forecasting through clinical data evaluation. Necessary pre-processing, including missing value filling, scaling, and feature selection, has been performed on the dataset. Multiple models, including Decision Tree, Random Forest, Logistic Regression, and K-Nearest Neighbors, have been developed and analyzed. A Soft Voting Ensemble Model has been introduced by combining the most effective models. The proposed model achieved 97.06% accuracy, which is better than single models and recent studies. The main reason for this success is the combination of multiple algorithms to reduce bias, reduce errors, and increase the reliability of predictions.**

Keywords*—Machine learning, ensemble models, prostate cancer, prediction model, soft-voting ensemble model.

I. Introduction

Prostate cancer (PCa) is one of the most common and deadly cancers in men globally. According to the World Health Organization (WHO), approximately 1.4 million new cases and more than 370,000 deaths were reported worldwide in 2020 alone **[1].** Prostate glands produce around 20% of seminal fluid, and unhealthy prostates impact urination, ejaculation, and bowel movements. Timely and appropriate differential diagnosis is very important in the diagnosis of prostate cancer, as it greatly reduces mortality and increases the chances of successful treatment of the patient. But the issue is that early symptoms of PCa often overlap with those of benign prostatic hyperplasia (BPH) and other urological problems. For example: difficulty urinating, especially frequent urination at night, weak urine stream, sexual dysfunction, blood in the urine, or pain in the waist and bones. Because of the similarity of these symptoms, it becomes difficult to reliably detect in early stages through traditional clinical methods **[1]**.

Commonly used methods in the medical field for early detection of prostate cancer (PCa) include biopsy **[2]**, and prostate-specific antigen (PSA) test **[3,4].**

In this study, we propose two new ensemble models for prostate cancer (PCa) diagnosis: Soft-Voting Ensemble Model (SVEM) and Hard-Voting Ensemble Model (HVEM). These models use a voting mechanism to combine predictions from various machine learning algorithms (e.g., KNN, Random Forest (RF), Logistic Regression (LR), and Support Vector Machine (SVM)). We assessed the models on a PCa dataset split into a 66:34 train-test ratio. The results show that the proposed SVEM models obtain a maximum accuracy of 97.06%, which is higher than the previous best-performing single model of 91.00%. This study underscores the need for initial steps to enhance model reliability.

- **Comprehensive Data Preprocessing:** Data balance and quality are ensured using SelectKBest, Min–Max scaling, SMOTEN and Hyperparameter Tuning.
- **Model Analysis:** KNN, Neural Network, Logistic Regression, SVM, and various tree-based and boosting models are analyzed for optimal performance.
- **Ensemble Learning:** Model stability and prediction reliability are increased by implementing Hard and Soft Voting Ensemble.
- **In-depth Performance Evaluation:** Soft Voting Ensemble achieved 97.06% accuracy, which was verified by Precision, Recall, F1-score, Confusion Matrix and ROC Curve.

This study is divided into four main sections. The Literature Review section provides a short overview of past studies in the related field. The Materials and Methods section outline the dataset and approach applied. The Results and Discussion section interprets the experimental results, and the Conclusion section highlights key findings and recommends directions for future work.

II. Literature Review

Due to the complexity of medical data collection and the sensitivity of patient information, obtaining high-quality prostate cancer (PCa) datasets remains a challenge. Most studies rely on publicly available or shared datasets under specific contracts. To our knowledge, Mahadi et al. **[5]** have

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achieved state-of-the-art results by applying machine learning models with GridSearchCV on the "PCa" dataset. In their study, LGBM, XGBoost and Random Forest models obtained a maximum accuracy of 96.67%. This shows that hyperparameter optimization via GridSearchCV is very important for boosting the performance of machine learning models.

Hasan et al. **[6]** studied prostate cancer prediction using six different machine learning methods, including Random Forest, SVM, KNN, Logistic Regression, Neural Networks, and Ensemble model. Using 100 patient records and 10 clinical features from the Kaggle dataset, they showed that Logistic Regression, Neural Networks, and Ensemble models achieved a maximum accuracy of 95%. In particular, the Ensemble method achieved a true positive detection rate of 96.55%, which is better than SVM and Random Forest (85%). Muktevi Srivenkatesh **[7]** proposed a machine learning-based approach for prostate cancer prediction using the Kaggle dataset. After solving missing values, they compared Logistic Regression, Random Forest, Naive Bayes, and SVM models. Their research demonstrates that Logistic Regression and Random Forest achieved the maximum accuracy of 90%, which highlights the role of simple ML algorithms in prostate cancer prediction. Cao-Van et al. **[8]** put forward a Soft-Voting Ensemble Model (SVEM) that integrates Logistic Regression, Random Forest, XGBoost, LGBM and SVM models for prostate cancer classification. In the experiment with the Kaggle dataset split in a ratio of 67:33, SVEM achieved a maximum accuracy of 96.97%, which outperformed the single models and showed that ensemble voting reduces errors and increases the reliability of predictions.

SLabidi and Issaoui **[9]** have tested a number of machine learning techniques, such as including recurrent neural networks (RNN), logistic regression (LR), naive Bayes, support vector machines (SVM), random forests (RF), decision trees (DT), K-nearest neighbors (KNN), and XGBoost. These models were tested on both normalized and raw datasets and evaluated using metrics such as accuracy, AUC, and F1-score. Their RNN model secured the highest results with 81% accuracy, AUC of 0.866 and F1 score of 0.809. Oyewo and Boyinbode **[10]** highlighted the importance of ensemble models to boost diagnostic precision through the integration of models such as SVM, DT, and Multilayer Perceptron (MLP). Their ensemble model obtained fascinating results: 99.06% accuracy, 98.09% sensitivity, and 99.54% specificity, greatly exceeding the performance of individual models such as LR (85.09%) and RF (83.10%). Wang and Jin **[11]** utilized the MIMIC-IV dataset and carried out thorough preprocessing which involved managing missing data, identifying outliers with isolation forests, and performing recursive feature elimination (RFE). Among the evaluated models (LR, SVM, DNN, XGBoost, LGBM, CatBoost), LGBM delivered the highest performance, attaining an AUC of 0.93. In addition to structured data, methods based on imaging have become popular. Handling imbalanced data has also been prominently exhibited in the work of Beltozer-Clemente et al. **[12]**, who analyzed a Kaggle dataset of 734 patients. After discovering numerous missing values, they utilized decision tree regressor and classifier models to predict and complete the absent entries. They performed data balancing using SMOTE and employed RFE for feature selection. Their experiments involving gradient boosting, RF, DT, and AdaBoost revealed that AdaBoost achieved the highest accuracy of 99.21%.

III. Materials and Methodology

The dataset that was obtained from Kaggle was carefully selected and prepared as necessary. We checked and ensured that the dataset does not contain any null values. We removed the ID column from the dataset. We used SMOTEN to oversample the unbalanced data. Different feature selection algorithms were involved in our research methodology. By using SelectKBest, we got a better score. By rating them according to their scores, we chose the top 5 features for our study. We segmented the dataset into training data and test data after feature selection. We used various machine learning and deep learning models to train the data. After running machine learning and deep learning models, we performed voting and stacking with the best performing model to get a better score. Our aim is to evaluate each model using key performance metrics like accuracy, precision, F1 score, and ROC score. Fig. 1 illustrates our methodology's working process.

A. Dataset

The dataset we used for this research is collected from Kaggle open-source Repository [**13**]. There are 100 instances and 10 features such as radius, texture, perimeter, area, fractal_dimension in the dataset. These features represent the clinical and biological indicators relevant to diagnosis. The target column diagnosis_result has two values which are Benign (B) and Malignant (M). Malignant means that this patient has prostate cancer, and Benign means the patient is safe from having prostate cancer.

B. Data Preprocessing

First of all, the id column is removed from the dataset as it is not related to the target column. We checked and ensured that the dataset does not contain any null values. In our study, the dataset was divided into two parts, where 66% of the data was used for training and 34% for testing. This dataset contains 62 instances, which are labelled as "B" and 38 instances, which are labelled as "M". Applying machine learning and deep learning models in this dataset can cause overfitting. We applied SMOTEN to remove overfitting by oversampling the minority class. We have used Min-Max scaling to scale all feature values from 0 to 1. So, it eliminates the possibility of a single feature dominating the model. We applied label encoding to convert the values of our target column from categorical to numerical to make machine learning models compatible.

C. Feature Selection

Feature selection is an effective way to improve accuracy and reduce learning time by removing redundant and irrelevant features [14]. We have applied various feature

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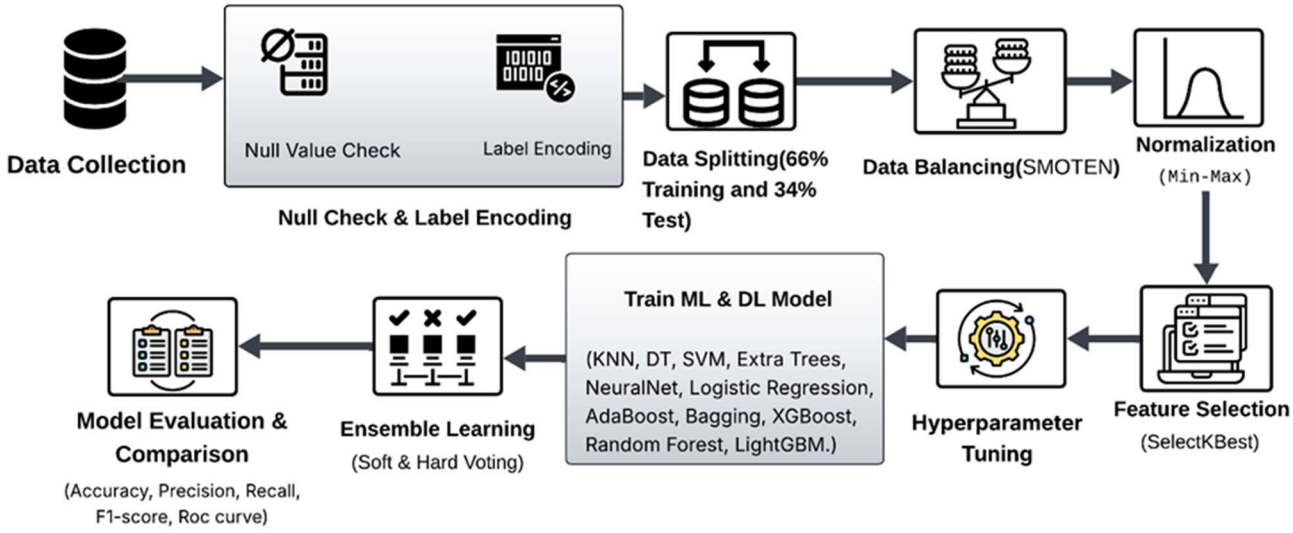


Fig. 1. Flow of Work Execution

selection methods like SelectKBest, MI, XGBoost, and Chi-Square. We get the highest score by using the SelectKBest method from Scikit-learn. We used median feature score as a threshold and only features with scores greater than or equal to the median were retained for subsequent model training.

D. Machine Learning and Deep Learning Models

As a key segment of this research, we have used a range of machine learning and deep learning techniques to predict prostate cancer, in order to predict prostate cancer accurately.

We have used a range of machine learning and deep learning models such as KNN, DT, SVM, Extra Trees, NeuralNet, Logistic Regression, AdaBoost, Bagging, XGBoost, Random Forest, LightGBM. Proper hyperparameters were set before model training for improved results.

E. Ensemble Learning Techniques

We have applied two ensemble techniques: voting and stacking to get better and more accurate accuracy. Usually, both hard voting and soft voting deliver more accurate results than individual models **[15]**. We have applied both hard and soft voting to combine predictions from 2-3 models for each possible combination. We applied stacking to make the model even more accurate. In stacking, multiple models make predictions and another model named meta model learns the best way to combine these predictions into the final prediction.

IV. Experimental Results and Discussion

This section presents experimental results and analyzes research findings. Through the analysis of the results, the relevance of the hypotheses presented in the study is assessed, and the significance and implications of the results are discussed.

A. Experimental Equipment

The tests were conducted on a computer with Windows 10 Home, Intel Core i5-1135G7 @ 2.40 GHz processor, and 8 GB RAM. Hyperparameters were kept the same across all models for fair and unbiased comparison.

B. Performance Evaluation

The confusion matrix is an important analytical tool for evaluating the performance of classification models in machine learning.

True Positive (TP): When both the actual and predicted outcomes are positive.

False Positive (FP): When the model predicts positive, the actual outcome is negative.

True Negative (TN): When both the predicted and actual outcomes are negative.

False Negative (FN): When the actual outcome is positive, the model predicts negative.

Based on the above values, the following performance indicators have been determined: equation (1), (2), (3), and (4).

$$\text{Accuracy} = \frac{TP+TN}{Total\ Sample} \quad (1)$$

This expresses the proportion of correct predictions out of the total predictions.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

This indicates how many positive predictions are correct out of the total number of predictions made.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (3)$$

It expresses how many of the true positive events the model correctly identified.

$$\text{F1 Score} = \frac{2\times Precision\times Recall}{Precision+Recal} \quad (4)$$

It maintains a balance between precision and recall and is useful for analyzing uneven datasets.

These are performance metrics to evaluate the performance of the prostate cancer prediction model. Accuracy shows the overall accuracy, while Precision and Recall evaluate positive and negative classification. The F1 score, combined with these, helps improve the model for more accurate disease detection.

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C. Results of Our Research

In this section, we have discussed in detail the development of various machine learning models such as XGBoost, RF, LGBM, SVM, and LR. Initially, 34% of the dataset was used for testing and the remaining 66% for training. Some limited results were obtained by applying only the basic model, which was improved in the next step through hyperparameter tuning and voting-based ensemble methods. The models achieved optimal performance by completing multiple training iterations. The overall results of this experiment are presented in Table I.

TABLE I. Performance of Single Classification Models

Models	Accuracy (%)	Precision (%)	Recall	F1-Score (%)	Running Time
RF	82.00	83.00	82.00	83.00	0.192s
LightGBM	82.00	85.00	82.00	83.00	0.020s
XGBoost	82.00	82.00	82.00	82.00	0.212s
Bagging	82.00	83.00	82.00	83.00	0.034s
AdaBoost	82.00	83.00	82.00	83.00	0.146s
Logistic Regression	85.00	86.00	85.00	85.00	0.009s
NeuralNet(MLP)	88.00	88.00	88.00	88.00	0.003s
Extra Trees	88.00	89.00	88.00	88.00	0.181s
SVM	88.00	89.00	88.00	88.00	0.003s
DT	88.00	89.00	88.00	88.00	0.003s
KNN	91.00	91.00	91.00	91.00	0.002s

From Table I, it can be seen that all five models achieved significant accuracy, where KNN achieved the highest accuracy of 91.00%. Other models such as NeuralNet (MLP), SVM, DT and Extra Trees achieved 88.00% accuracy respectively. Logistic Regression achieved 85.00% accuracy. They demonstrated good classification performance due to their high Precision and Recall values. The F1-score also shows that there was a good balance between Precision and Recall. Almost all models completed training in a very short time (below 0.2 seconds).

We did not limit our work to implementing the models, the accuracy was further improved by hyperparameter tuning of the models and various voting ensemble techniques. In the next step of the research, ensemble learning was applied by combining multiple single models with hard and soft voting methods. A summary of these ensemble models is presented in Table II.

Analyzing Table II, it can be seen that the models combined with the soft voting method showed significant improvement in data classification. The KNN + RF, NeuralNet + RF, and LR + RF soft voting combinations achieved 97.06% accuracy, and almost all Precision, Recall, and F1-score values were close to 1.00. KNN + RF, SVM + RF, NeuralNet + RF, and LR + RF achieved the same high level of accuracy in the hard voting method. While the single models achieved a maximum accuracy of 91.00% in the first stage, this accuracy was increased to 97.06% through voting and tuning. The training time of the ensemble models was also quite tolerable, which makes them applicable in real environments. The training time of the soft voting combinations ranged from 0.149 seconds to 0.348 seconds, and that of the hard voting combinations ranged from 0.005 seconds to 0.216 seconds

One of the main reasons for this performance increase through model combination is that each single model has its own prediction limitations, which complement each other when used together. The soft voting method uses the average of the prediction probabilities of each model to determine the final class, thus reducing common errors or bias and allowing correct conclusions to be reached.

TABLE II. Experimental Results of Ensemble Machine Learning Models.

Voting		Accuracy (%)	Precision (%)	Recall	F1-Score	Running Time
Hard Voting	SVM+KNN	94.12	0.94	0.94	0.94	0.005s
	LR+KNN	94.12	1.00	0.94	0.94	0.006s
	LR+NeuralNet	94.12	1.00	0.94	0.94	0.209s
	LR+SVM	94.12	1.00	0.94	0.94	0.007s
	KNN+RF	97.06	1.00	0.94	0.97	0.151s
	SVM+RF	97.06	1.00	0.94	0.96	0.116s
	NeuralNet+RF	97.06	1.00	0.94	0.97	0.348s
	LR+RF	97.06	1.00	0.94	0.97	0.216s
Soft Voting	LR+NeuralNet + SVM	94.12	0.94	0.94	0.94	0.136s
	NeuralNet+KNN	94.12	0.94	0.94	0.94	0.179s
	LR+SVM	94.12	1.00	0.94	0.94	0.009s
	LR+SVM+RF	97.06	1.00	0.94	0.97	0.202s
	LR+KNN+RF	97.06	1.00	0.94	0.97	0.213s
	NeuralNet+SVM+RF	97.06	1.00	0.94	0.97	0.317s
	KNN+RF	97.06	1.00	0.94	0.97	0.151s
	SVM+RF	97.06	1.00	0.94	0.96	0.149s
	NeuralNet+RF	97.06	1.00	0.94	0.97	0.325s
	LR+RF	97.06	1.00	0.94	0.97	0.206s

Finally, the effectiveness of multiple single models has been verified by initially testing with 34% of the dataset and training with 66%. In the next step, the performance on PCa data has been significantly improved by applying hyperparameter tuning and ensemble learning. This confirms the effectiveness and practicality of the ensemble learning method in medical diagnostic work.

D. Comparison of Our Results with Existing Paper

In this study, a comparative analysis was conducted with the results of various previous studies to verify the effectiveness of the proposed models. Four evaluation indices — Accuracy, Precision, Recall, and F1-score — were used in this analysis. Table III presents the results of our proposed models and existing studies, which provides a comprehensive and organized framework for evaluating effectiveness.

Notably, our proposed multiple ensemble models namely, KNN+RF, NeuralNet+RF, and LR+RF achieved a maximum accuracy of 97.06%, which is better than the results of all previous studies. In comparison, the study by Kien Cao-Van et al. **[8]** showed a maximum accuracy of 96.97%,

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M.Srivenkatesh et al. **[7]** achieved 90.00% and the cases of Hasan et al. **[6]** were 95.00%. and M. K. Mahadi et al. **[5]** were 96.67%. Almost all of our models achieved a value of 1.00 in the precision index, which is better than most previous studies. For example, Hasan et al. **[6].** The precision of its Ensemble model was 0.96 and that of Mahadi's LGBM model was 0.92. Our models simultaneously achieved high recall (0.94) and F1-score (maximum 0.97), indicating the balanced classification performance of the models.

TABLE III. Comparison with Previous Studies

Research	**Model**	**Accuracy (%)**	**Precission**	**Recall**	**F1-Score (%)**
Our Model	KNN+RF	97.06	1.00	0.94	0.97
	SVM+RF	97.06	1.00	0.94	0.96
	NeuralNet+RF	97.06	1.00	0.94	0.97
	LR+RF	97.06	1.00	0.94	0.97
K.Cao Van et al. **[8]**	XGBoost+RF	96.97	1.00	0.96	0.98
	RF +LGBM	96.97	1.00	0.96	0.98
	RF+LR	96.97	1.00	0.96	0.98
M.Srivenkatesh et al. **[7]**	KNN	80.00	1.00	0.67	0.80
	RF	90.00	1.00	0.83	0.91
	LR	90.00	1.00	0.83	0.91
S. M. Hasan et al. **[6]**	KNN	90.00	0.88	0.89	0.90
	NeuralNet	95.00	0.93	0.87	0.90
	LR	95.00	0.96	0.95	0.95
	Ensemble Model	95.00	0.96	0.95	0.95
M.K. Mahadi et al. [**5**]	XGBoost	96.67	1.00	0.91	0.95
	RF	96.67	1.00	0.91	0.95
	LGBM	96.67	0.92	1.00	0.96

Although some of Kien Cao-Van's **[8]** models achieved F1-scores as high as 0.98 (e.g., XGBoost+RF, RF+LGBM, RF+LR), our models showed similar or relatively close performance and were able to maintain this high performance across a larger number of models.

From the comparative analysis results, it is clear that our proposed methods not only achieved high accuracy but also demonstrated a stable and reliable performance in terms of precision, recall, and F1-score. In addition, the computational efficiency and fast training time of the models make them suitable for application on large datasets.

Overall, this comparative analysis indicates that our proposed models are capable of being useful in real-life applications as well as providing a solid foundation for future research.

Fig. 2 shows the confusion matrix of proposed Voting Classifier performed very well. Out of a total of 34 test samples, 33 were correctly classified. Specifically, the model correctly identified 17 class 0 samples (True Negative) and 16 class 1 samples (True Positive). Only one misclassification occurred, where a class 1 sample was predicted as class 0 (False Negative). Notably, no false positives were found, indicating 100% precision for class 0 (This is one within six voting models).

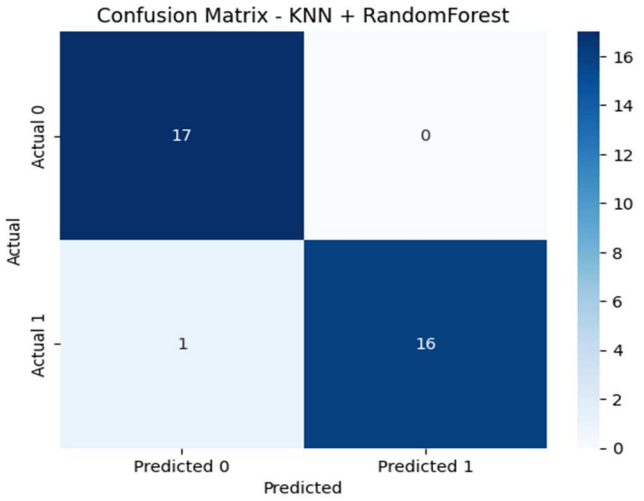


Fig. 2. Confusion Matrix of the Soft Voting Classifier (KNN + Random Forest).

The ROC curve shown in Fig. 3 further confirms the model's performance. The curve almost completely reaches the ideal upper-left corner, and the AUC value is found to be 1.00. This proves that the Voting Classifier is almost perfectly capable of separating the two classes.

Fig. 3 shows that Random Forest, LightGBM, XGBoost, Bagging and AdaBoost showed relatively stable but limited discrimination power (Accuracy around 82%). Logistic Regression achieved slightly better results (85%). On the other hand, SVM, Decision Tree, Extra Trees, and Neural Network (MLP) achieved higher accuracy (88%) and their ROC–AUC values were relatively large. Although MLP performed well, its ROC curve was slightly unstable, indicating sensitivity to threshold changes. Among all the models, K-Nearest Neighbors (KNN) showed the highest performance (Accuracy 91%), and its ROC–AUC value was the highest according to Fig. 3. This proves that KNN is able to classify reliably across different thresholds.

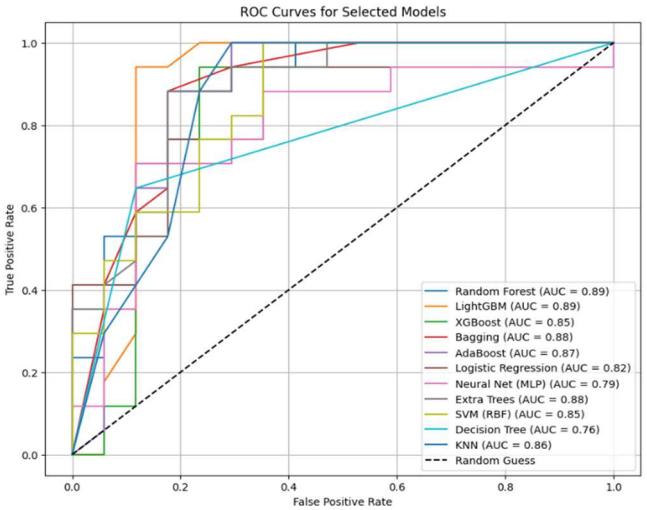


Fig. 3. ROC Curve for All Single Classifier Models.

Overall, the ROC–AUC analysis shown in Fig. 3 is consistent with the results in the table, and its use alongside conventional metrics is important in model evaluation. The results indicate that KNN is the most effective classifier for

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this dataset, although SVM, Decision Tree, and Extra Trees demonstrated competitive performance.

V. Conclusion

This study proposes an advanced ensemble machine learning framework for prostate cancer prediction using clinical data. The study focuses on data preparation, such as missing data handling, feature selection, data balancing (SMOTEN), and scaling, which ensure data quality and model stability. Here, various machine learning models such as KNN, SVM, Logistic Regression, Random Forest, and Neural Network are implemented separately, where the highest accuracy of 91.00% is obtained. Later, the power of the models is combined by applying hard voting and soft voting-based ensemble techniques. The results show that the proposed ensemble model achieves a maximum accuracy of 97.06%, which is better than the single model and existing studies. In addition, the results of high precision, recall, and F1-score confirm its effectiveness and stability. These results prove that the ensemble method reduces errors and is capable of providing accurate predictions. Overall, this study can help physicians with early and reliable detection of prostate cancer and lays out a solid foundation for future large dataset-based research. Future studies will validate the effectiveness of the proposed framework using larger and multicenter clinical datasets and incorporate Explainable AI techniques to enhance the model's clarity, generalizability, and real-world clinical applicability.

VI. Reference

[1] B. A. Akinnuwesi, K. A. Olayanju, B. S. Aribisala, S. G. Fashoto, E. Mbunge, M. Okpeku, and P. Owate, "Application of support vector machine algorithm for early differential diagnosis of prostate cancer," *Data Science and Management*, vol. 6, no. 1, pp. 1–12, 2023, doi: 10.1016/j.dsm.2022.10.001

[2] E. Benedetti, K. Chetnik, T. Flynn, C. E. Barbieri, D. S. Scherr, M. Loda, and J. Krumsiek, "Plasma metabolomics profiling of 580 patients from an Early Detection Research Network prostate cancer cohort," *Scientific Data*, vol. 10, no. 1, p. 830, 2023, doi: 10.21228/M86H7K.

[3] T. Nordström, O. Akre, M. Aly, H. Grönberg, and M. Eklund, "Prostate-specific antigen (PSA) density in the diagnostic algorithm of prostate cancer," *Prostate Cancer and Prostatic Diseases*, vol. 21, no. 1, pp. 57–63, 2018, doi: 10.1016/S0140-6736(14)60525-0.

[4] J. R. Prensner, M. A. Rubin, J. T. Wei, and A. M. Chinnaiyan, "Beyond PSA: the next generation of prostate cancer biomarkers," *Science Translational Medicine*, vol. 4, no. 127, pp. 127rv3, 2012, doi: 10.1126/scitranslmed.3003180.

[5] M. K. Mahadi, S. R. Abir, A. M. Moon, M. Adnan, M. A. N. I. Khan, M. M. Nishat, et al., "Machine learning assisted decision support system for prediction of prostate cancer," in *Proc. 2023 20th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*, May 2023, pp. 1–5, doi: 10.1109/ECTI-CON58255.2023.10153167.

[6] S. M. Hasan, M. F. Rabbi, and N. Jahan, "Can machine learning technique predict the prostate cancer accurately?: The fact and remedy," in *Proc. 2021 Int. Conf. Electronics, Communications and Information Technology (ICECIT)*, Sept. 2021, pp. 1–4, doi: 10.1109/ICECIT54077.2021.9641099.

[7] M. Srivenkatesh, "Prediction of prostate cancer using machine learning algorithms," *International Journal of Recent Technology and Engineering (IJRTE)*, vol. 8, no. 5, pp. 5353–5362, 2020, doi: 10.35940/ijrte.E6754.018520.

[8] K. Cao-Van, L. G. M. Tran Cao Minh, T. B. Q. Tran Thị, and H. M. Tan, "Soft-Voting Ensemble Model: An Efficient Learning Approach for Predictive Prostate Cancer Risk," *Vietnam Journal of Computer Science*, vol. 11, no. 4, pp. 531–552, Aug. 2024.

[9] A. Laabidi and M. Aissaoui, "Performance analysis of machine learning classifiers for predicting diabetes and prostate cancer," in *Proc. 2020 1st International Conference on Innovative Research in Applied Science, Engineering and Technology (IRASET)*, Meknes, Morocco, 2020, pp. 1–6, doi: 10.1109/IRASET48871.2020.9092255.

[10] O. A. Oyewo and O. K. Boyinbode, "Prediction of prostate cancer using ensemble of machine learning techniques," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 3, 2020, doi: 10.14569/IJACSA.2020.0110318.

[11] W. Wang and X. Jin, "Prostate cancer prediction model: A retrospective analysis based on machine learning using the MIMIC-IV database," *Intelligent Pharmacy*, vol. 1, no. 4, pp. 268–273, 2023, doi: 10.1016/j.ipha.2023.04.010.

[12] S. Beltozar-Clemente, E. Diaz-Vega, I. C. Ramos, and R. T. Navarrete, "Improving accuracy: Comparative analysis of machine learning models for prostate cancer prediction," *International Journal of Intelligent Systems Applications in Engineering*, vol. 12, no. 2, pp. 654–664, 2024.

[13] Sajidsaifi. *Prostate Cancer*. Kaggle, 2025. Available online: https://www.kaggle.com/datasets/sajidsaifi/prostate-cancer (accessed on 07 August 2025).

[14] J. Cai, J. Luo, S. Wang, and S. Yang, "Feature selection in machine learning: A new perspective," *Neurocomputing*, vol. 300, pp. 70–79, 2018, doi: 10.1016/j.neucom.2017.11.077.

[15] N. Peppes, E. Daskalakis, T. Alexakis, E. Adamopoulou, and K. Demestichas, "Performance of machine learning-based multi-model voting ensemble methods for network threat detection in agriculture 4.0," *Sensors*, vol. 21, no. 22, p. 7475, 2021, doi: 10.3390/s21227475.

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